Analysis of Feedback

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1 Introduction

- We are interested in the closed-loop system properties that emerge from feedback-control:
  - The behavior of the feedback controlled system can be completely different from that of the open-loop system, and under the engineering control of the controls engineer.
  - The controls engineer is often interested in four categories of system performance specification:
    1. Stability and damping
       * In almost all cases, the closed-loop system must be stable to be useful, and often good damping is required.
    2. Accuracy (tracking)
       * Many systems must meet an accuracy specification
    3. Speed (regulation)
       * Many systems must meet an speed specification
    4. Sensitivity and robustness
       * The system should not be highly sensitive to variations in the plant transfer function.
- In this part you will learn to evaluate
  - Accuracy:
    * Calculating steady-state errors
    * Plotting tracking errors
  - Determine the sensitivity transfer function

2 Example case: motor drive

- We consider a case study to illustrate the four categories of system performance.

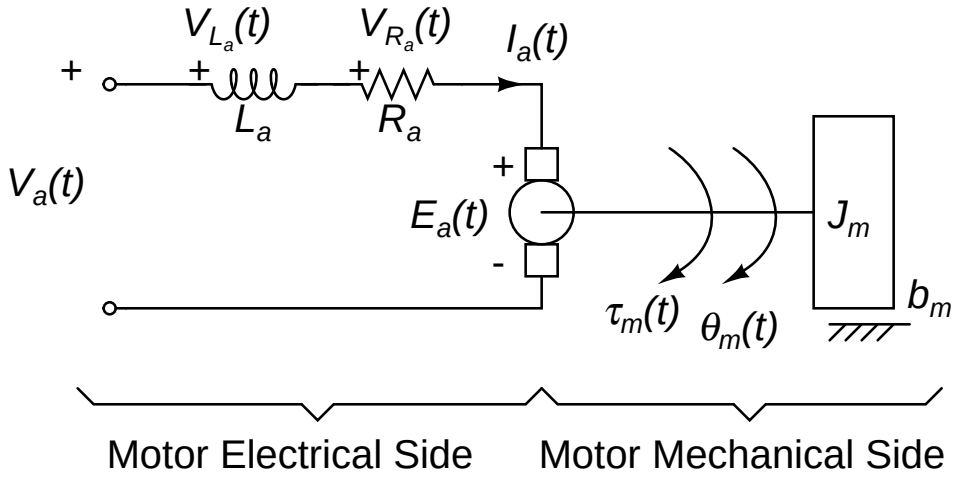


Figure 1: Schematic diagram of motor drive.

K_m	Motor constant	$\left[\frac{\text{N-m}}{\text{Amp}}\right]$ $\left[\frac{\text{Volts}}{\text{rad/sec}}\right]$	0.1537
J_m	Inertial	$\left[\frac{\text{N-m-}s^2}{\text{radian}}\right]$	7.56×10^{-4}
b_m	Damping (friction)	$\left[\frac{\text{N-m-s}}{\text{radian}}\right]$	0.0034
R_a	Armature resistance	$\left[\frac{\text{volts}}{\text{amp}}\right]$	7.2
L_a	Armature Inductance	$\left[\frac{\text{volt-seconds}}{\text{amp}}\right]$	0.145

Table 1: Parameters of the motor drive.

$V_a(t)$	Applied voltage	[volts]
$I_a(t)$	Armature current	[amps]
$\tau_m(t)$	Motor torque	[N-m]
$\theta(t)$	Motor position	[radians]

Table 2: Signals of the motor drive.

2.1 Transfer function and open-loop response

$$\text{Voltage -> velocity :} \quad G_{uv}(s) = \frac{K_m}{J_m L_a s^2 + (J_m R_a + b_m L_a)\, s + (b_m R_a + K_m^2)}$$
$$= \frac{0.1537}{0.0001096\, s^2 + 0.00593\, s + +0.04782}$$
$$= \frac{1402}{s^2 + 54.1\, s + 436.2} \tag{1}$$

$$\text{Voltage -> position :} \quad G_{uy}(s) = \frac{K_m}{J_m L_a s^2 + (J_m R_a + b_m L_a)\, s + (b_m R_a + K_m^2)} \frac{1}{s}$$
$$= \frac{1402}{s^3 + 54.1\, s^2 + 436.2\, s} \tag{2}$$

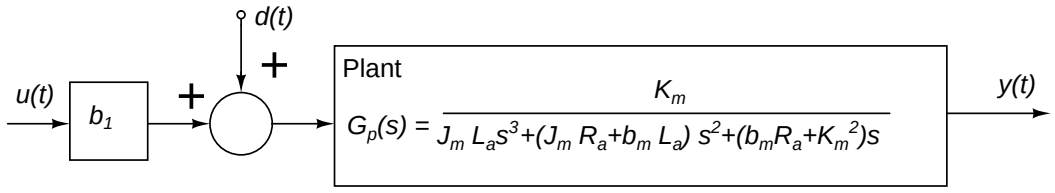
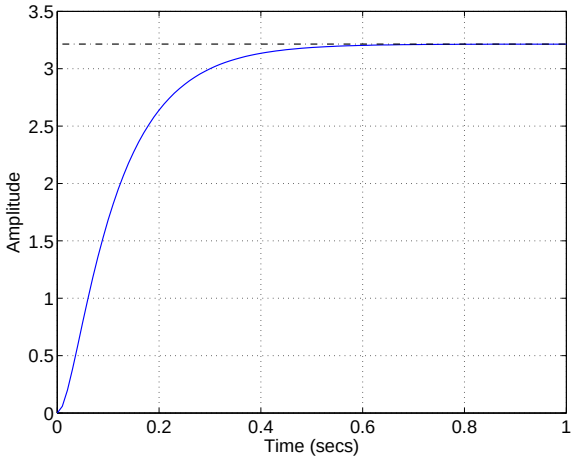
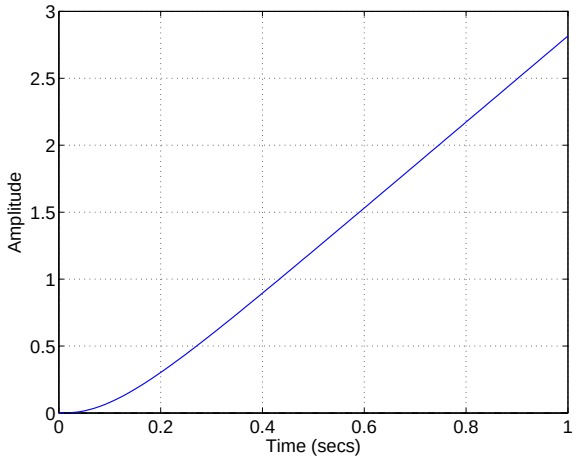


Figure 2: Open-loop block diagram of motor drive.



a) Velocity, `>>step(Guv)`



b) Position, `>>step(Guy)`

Figure 3: Open-loop step response, velocity and position .

3 Calculating steady-state errors (Franklin *et al.* sec. 4.2 (7th ed.))

- Steady-state errors are computed from the transfer function $r(t) \rightarrow e(t)$, and the final value theorem:

  Final value theorem: For a stable system with transfer function $T_{re}(s) = E(s)/R(s)$,

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) \tag{3}$$

  In Eqn (3) $E(s)$ (a Laplace transform) is given by $T_{re}(s)$ (a transfer function) times $R(s)$ (a Laplace transform):

$$E(s) = T_{re}(s)\,R(s) \tag{4}$$

  which gives

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sT_{re}(s)\,R(s) \tag{5}$$

  – To compute the final value of the error, we have to specify the input $r(t)$. Signals typically used to compute steady-state error are seen in table 3.

The step, ramp and acceleration input signals and the Laplace Transforms can be written:

$$r(t) = \frac{1}{k!}t^k 1^+(t)\ , \qquad R(s) = \frac{1}{s^{k+1}}\ . \tag{6}$$

For $k = 1,2,3$ the signals and Laplace transforms are given in table 3.

	k	Test input signal, $r(t)$	$R(s)$
Step input	0	$r(t) = 1^+(t)$	$R(s) = \frac{1}{s}$
Ramp input	1	$r(t) = t\,1^+(t)$	$R(s) = \frac{1}{s^2}$
Accelerating input	2	$r(t) = \frac{1}{2}t^2\,1^+(t)$	$R(s) = \frac{1}{s^3}$

Table 3: Typical inputs to consider for determining steady-state error.

3.1 Steady-state error example

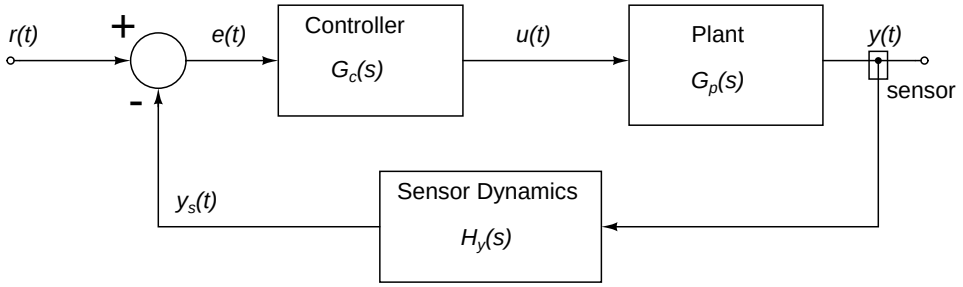


Figure 4: Standard block diagram.

- From the standard block diagram, figure 4, the error transfer function is given as:

$$T_{re}(s) = \frac{1}{1 + G_c(s)\, G_p(s)\, H_y(s)} \tag{7}$$

- Considering the velocity-controlled motor example of section 2.1 with proportional control:

$$\begin{aligned} G_c(s) &= k_p \\ G_p(s) &= G_{uv}(s) = \frac{1402}{s^2 + 54.1\, s + 436.2} \\ H_y(s) &= 1.0 \end{aligned}$$

- The error transfer function is given as:

$$T_{re}(s) = \frac{1}{1 + k_p \frac{1402}{s^2 + 54.1\, s + 436.2}} = \frac{s^2 + 54.1\, s + 436.2}{s^2 + 54.1\, s + 436.2 + k_p 1402}$$

- Plugging $T_{re}(s)$ into the final value theorem, and considering a step input:

$$e_{ss}(\text{step input}) = \lim_{s\to 0} s\,T_{re}(s)\,R(s) = \lim_{s\to 0} s\,\frac{s^2+54.1\,s+436.2}{s^2+54.1\,s+436.2+k_p\,1402}\,\frac{1}{s}$$

The s and $\frac{1}{s}$ cancel, to give

$$e_{ss}(\text{step input}) = \lim_{s\to 0} \frac{s^2+54.1\,s+436.2}{s^2+54.1\,s+436.2+k_p\,1402}$$

- Taking the limit removes all the higher powers of s

$$\begin{aligned} e_{ss}(\text{step input}) &= \frac{436.2}{436.2+k_p\,1402} \\ &= \frac{1}{1+3.21\,k_p} \end{aligned}$$

- Now consider a ramp input, $r(t) = t\,1^+(t)$, $R(s) = \frac{1}{s^2}$

$$e_{ss}(\text{ramp input}) = \lim_{s\to 0} s\,T_{re}(s)\,R(s) = \lim_{s\to 0} s\,\frac{s^2+54.1\,s+436.2}{s^2+54.1\,s+436.2+k_p\,1402}\,\frac{1}{s^2}$$

For the ramp input, only one of the $\frac{1}{s}$ terms cancels, giving

$$e_{ss} = \lim_{s\to 0} \frac{s^2+54.1\,s+436.2}{s^2+54.1\,s+436.2+k_p\,1402}\,\frac{1}{s} = \lim_{s\to 0} \frac{436.2}{436.2+k_p\,1402}\,\frac{1}{s} = \infty$$

- We see that for this system,

Input	Steady-state error
Step	$e_{ss} = \frac{1}{1+K_{Bode}}$
Ramp	$e_{ss} = \infty$

Table 4: Steady state errors for example case with proportional velocity feedback.

- The example shows that we can calculate e_{ss} directly from the final value theorem.

- However, it is possible to solve for the steady-state error in a general way by considering the loop gain in Bode standard form.

3.2 System type and Bode standard form

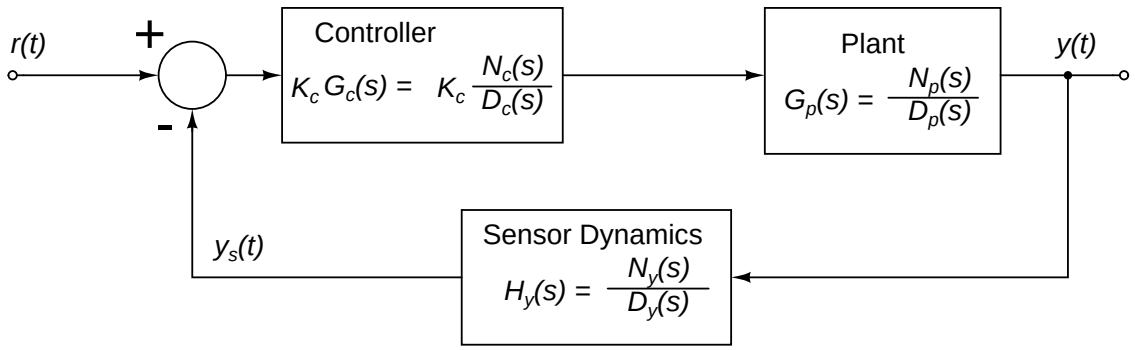


Figure 5: Standard feedback loop, with numerator and denominators called out.

- Considering figure 5, the error transfer function is given as:

$$T_{re}(s) = \frac{1}{1 + K_c G_c G_p H_y} = \frac{1}{1 + K_c \frac{N_c}{D_c}\frac{N_p}{D_p}\frac{N_y}{D_y}} = \frac{D_c D_p D_y}{D_c D_p D_y + K_c N_c N_p N_y}$$

- Defining the ***loop gain*** as $G_L(s) = G_c(s)\,G_p(s)\,H_y(s)$, the poles of the loop gain are the zeros of the error transfer function !
  - Therefore, if the loop gain has a pole at the origin, the error transfer function has a zero at the origin (which means its DC gain is zero)
  - If $T_{re}(s)$ has zeros at the origin, these cancel out $\frac{1}{s}$ terms coming from $R(s)$ in the final value theorem.
- The number of loop-gain poles at the origin is a sufficiently basic property of feedback systems that it is given a special name: the ***System Type***.
- The system type is given as

  Type 0: $G_L(s)$ has no pole at the origin, $N_o = 0$

  Type I: $G_L(s)$ has one pole at the origin, $N_o = 1$

  Type II: $G_L(s)$ has two poles at the origin, $N_o = 2$ etc.

  where N_o is the number of *open-loop* poles at the origin.

3.2.1 Determining steady-state error using Bode standard form

- Defining the loop gain as product of gains around the loop. For the standard loop (figure 4):

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) \tag{8}$$

- Writing the loop gain in Bode standard form:

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) = \frac{1}{s^{N_o}} K_{Bode} \frac{b_m s^m + b_{m-1} s^{m-1} \ldots + b_1 s + 1}{a_n s^n + a_{n-1} s^{n-1} \ldots a_1 s + 1} \tag{9}$$

We can write the error transfer function as:

$$T_{re}(s) = \frac{1}{1 + G_L(s)} \tag{10}$$

- Using Eqns (9) and (10), a general solution for the steady-state error can be found:

$$\begin{aligned} e_{ss} &= \lim_{s\to 0} s\, T_{re}(s)\, R(s) \\ &= \lim_{s\to 0} s \frac{1}{1 + \frac{1}{s^{N_o}} K_{Bode} \frac{b_m s^m + b_{m-1} s^{m-1} \ldots b_1 s + 1}{a_n s^n + a_{n-1} s^{n-1} + \ldots a_1 s + 1}} R(s) \end{aligned} \tag{11}$$

But in the limit, all of the terms $\left(b_m s^m + b_{m-1} s^{m-1} + .. + b_1 s\right)$ and $\left(a_m s^m + a_{m-1} s^{m-1} + .. + a_1 s\right)$ drop out, giving

$$\begin{aligned} e_{ss} &= \lim_{s\to 0} s \frac{1}{1 + \frac{1}{s^{N_o}} K_{Bode}} R(s) \\ &= \lim_{s\to 0} s \frac{s^{N_o}}{s^{N_o} + K_{Bode}} R(s) \end{aligned} \tag{12}$$

- Eqn (12) has a particularly simple form, and is valid for step, ramp or acceleration inputs, and any system type.

3.2.2 Example, position control of a motor drive

- Considering the position-controlled motor drive of section 2.1 with proportional control, the loop-gain is given as:

$$G_L(s) = G_c(s)\, G_p(s)\, H_y(s) \;=\; k_p \frac{1402}{s^3 + 54.1\, s^2 + 436.2\, s}$$

  Putting the loop-gain in standard Bode form:

$$G_L(s) \;=\; 3.21\, k_p \frac{1}{s^1} \frac{1}{0.0023\, s^2 + 0.124\, s + 1} \tag{13}$$

  - The position-controlled system is Type I, with $K_{Bode} = 3.21\, k_p$.

- Using Eqn (12)

$$e_{ss} = \lim_{s\to 0} s\, T_{re}(s)\, R(s) = \lim_{s\to 0} s \frac{s^{N_o}}{s^{N_o} + K_{Bode}} R(s) \tag{14}$$

  For a step input, $R(s) = \frac{1}{s}$ and $N_o = 1$ for the position-controlled motor gives

$$e_{ss}(\text{step input}) \;=\; \lim_{s\to 0} s\, T_{re}(s) \frac{1}{s} = \lim_{s\to 0} s \frac{s^1}{s^1 + K_{Bode}} \frac{1}{s} = \lim_{s\to 0} \frac{s}{K_{Bode}} = 0$$

- The steady-state error for a Type I system with a step input is 0.

- Considering a ramp input,

$$e_{ss}(\text{ramp input}) \;=\; \lim_{s\to 0} s\, T_{re}(s) \frac{1}{s^2} = \lim_{s\to 0} s \frac{s^1}{s^1 + K_{Bode}} \frac{1}{s^2} = \lim_{s\to 0} \frac{1}{K_{Bode}} = \frac{1}{K_{Bode}}$$

3.3 Table of steady-state errors

- Eqn (12) leads to a general form for the steady-state error. Recall

$$e_{ss} = \lim_{s\to 0} s \frac{s^{N_o}}{s^{N_o} + K_{Bode}} R(s) \tag{15}$$

- As we've seen in Eqn (6) and table 3,

$$R(s) = \frac{1}{s^{k+1}} \tag{16}$$

  where k indicates the step $(k = 0)$, ramp $(k = 1)$ or acceleration input $(k = 2)$.

- Inserting $R(s)$ from Eqn (16) into the equation for steady-state error gives

$$e_{ss} = \lim_{s\to 0} s \frac{s^{N_o}}{s^{N_o} + K_{Bode}} \frac{1}{s^{k+1}} \tag{17}$$

  where N_o is the system type, and k is the order of the input signal. Using Eqn (17), steady-state errors for system types 0, I and II are seen in table 5.

	k	$r(t)$	$R(s)$	Type 0 System	Type I System	Type II System
Step	0	$1^+(t)$	$R(s) = \frac{1}{s}$	$\frac{1}{1+K_{Bode}}$	0	0
Ramp	1	$t\,1^+(t)$	$R(s) = \frac{1}{s^2}$	∞	$\frac{1}{K_{Bode}}$	0
Accel	2	$\frac{1}{2}t^2\,1^+(t)$	$R(s) = \frac{1}{s^3}$	∞	∞	$\frac{1}{K_{Bode}}$

Table 5: Connection between system type and steady-state error.

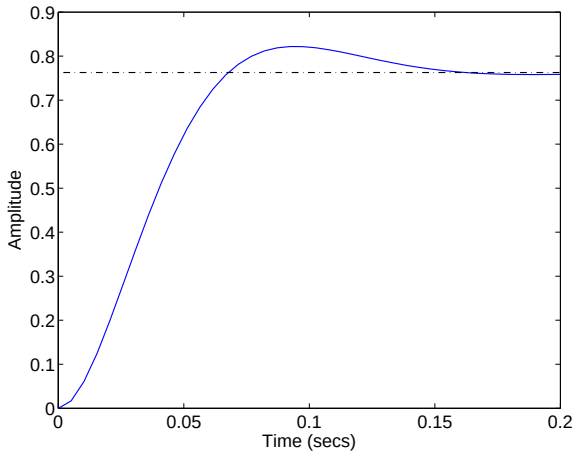
- Types 0 and I are the most common system types in practical systems. Type II is also seen. Type III and above is quite rare.

3.4 Summary for steady-state error

- The standard way to compute steady-state error is through table 5.
  - Table 5 shows that steady-state error depends on
    1. The system type,
    2. K_{Bode},
    3. The order of the input (step, ramp or acceleration).
- We often use the controller to make a Type 0 system into Type I, specifically to make the steady-state error go to zero for step inputs.
  - Controllers that increase the system type by one:
    * PI
    * PID
- K_{Bode} is the product of the DC gains around the loop. By increasing the controller gain we increase K_{Bode} and reduce steady-state errors.

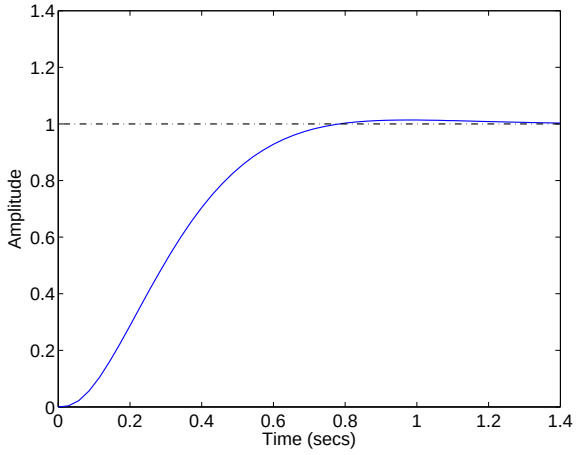
3.5 Example plots, steady-state error for the motor drive

- Proportional-type velocity control (system is type 0)



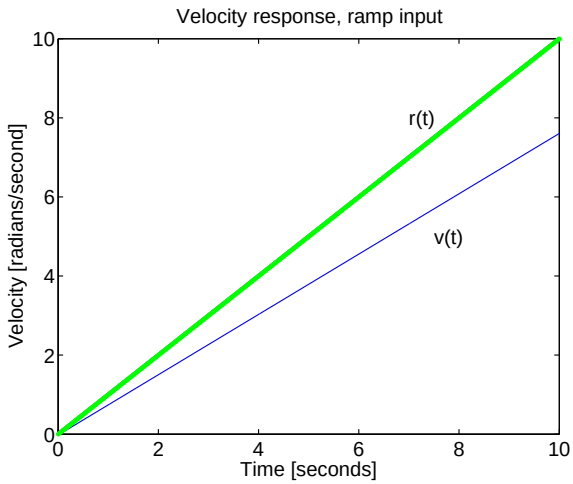

```
Guv = tf(1402, [1, 54.1, 346.2])
kp  = 1
Trv = feedback(kp*Guv, 1)
figure(5), clf
  step(Trv)
  SetLabels(12)
  print('-deps2c', 'Pcontrol_VelStep')
```

Figure 6: Step response of velocity, p-type control, system is type 0, SSE = $\frac{1}{1+K_{Bode}} = 0.237 = 24\%$.



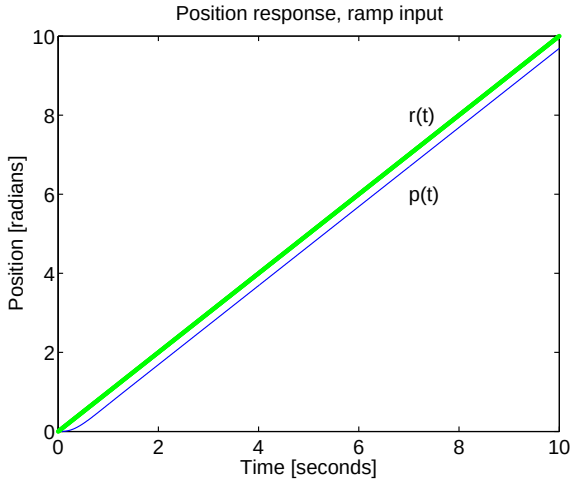

```
Guv = tf(1402, [1, 54.1, 346.2, 0])
Try = feedback(kp*Guy, 1)
figure(5), clf
  step(Try)
  SetLabels(12)
  print('-deps2c', 'Pcontrol_PosStep')
```

Figure 7: Step response of position, p-type control, system is type I, SSE = 0%.




```
rt = 0:0.01:10;
yt = lsim(Trv, rt, rt);
figure(6), clf
  plot(rt, yt, 'b-', rt, rt, 'g.-')
  SetLabels(12, 12,1)
  title('Velocity response, ramp input');
  xlabel('Time [seconds]');
  ylabel('Velocity [radians/second]');
  print('-deps2c', 'Pcontrol_VelRamp')
```

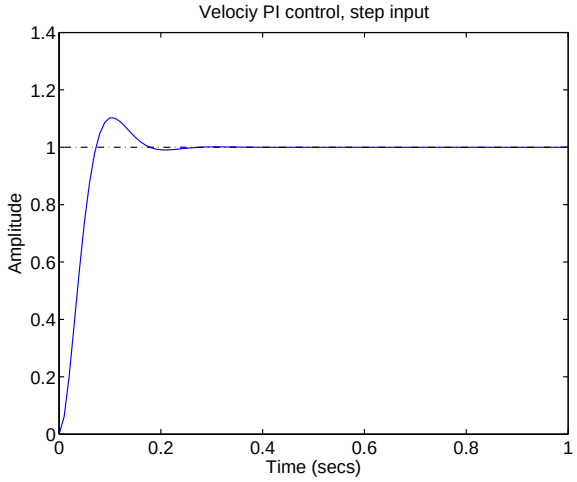
Figure 8: Ramp response of velocity, p-type control, system is type 0, SSE = ∞ .




```
yt = lsim(Try, rt, rt);
figure(6), clf
  plot(rt, yt, 'b-', rt, rt, 'g.-')
  SetLabels(12, 12,1)
  title('Position response, ramp input');
  xlabel('Time [seconds]');
  ylabel('Position [radians]');
  print('-deps2c', 'Pcontrol_PosRamp')
```

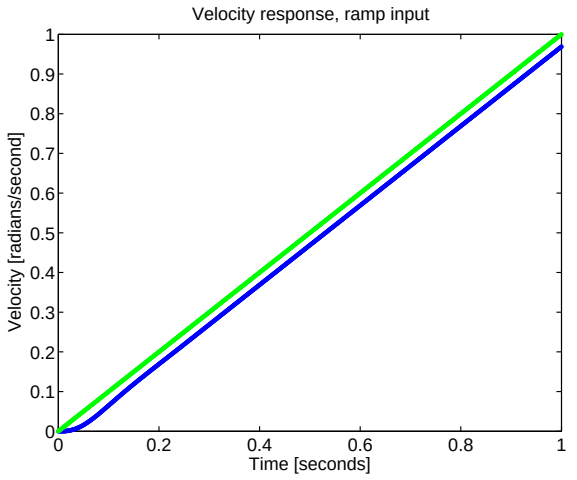
Figure 9: Ramp response of position, p-type control, system is type I, SSE=$\frac{1}{K_{Bode}} = 31\%$.

- Proportional-Integral (PI) control increases system type




```
kp = 1, ki = 10
Gc = tf([kp, ki], [1 0])
Try = feedback(Gc*Guv, Hy)
figure(7), clf
  step(Try, 1)
  title('Velociy PI control, step input')
  SetLabels(12, 12,1)
  print('-deps2c', 'PIcontrol_VelStep')
```

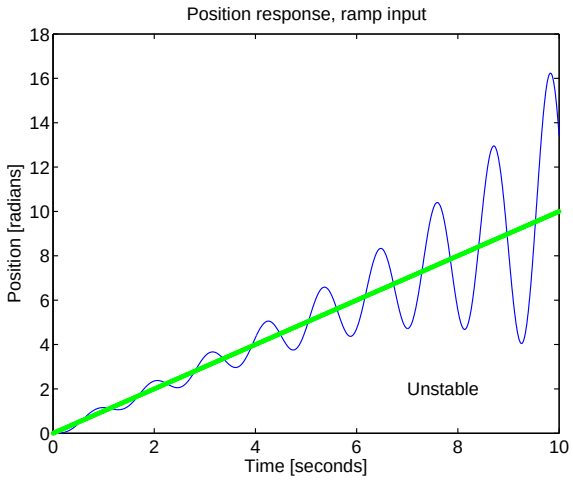
Figure 10: Step response of velocity with PI control, system is type I, SSE= 0.




```
Trv = feedback(Gc*Guv, Hy);
rt = 0:0.01:10;
yt = lsim(Trv, rt, rt);
figure(7), clf
  plot(rt, yt, 'b-', rt, rt, 'g.-')
  title('Velocity response, ramp input');
  xlabel('Time [seconds]');
  ylabel('Velocity [radians/second]');
  SetLabels(12, 12,1)
  print('-deps2c', 'PIcontrol_VelRamp')
```

Figure 11: Ramp response of velocity with PI control, system is type I, SSE= $\frac{1}{K_{Bode}} = 3.1\%$.

- But be careful, the motor drive is open-loop stable, but PI control can make it unstable !




```
Try = feedback(Gc*Guy, Hy);
yt = lsim(Try, rt, rt);
figure(7), clf
  plot(rt, yt, 'b-', rt, rt, 'g.-')
  title('Position response, ramp input');
  xlabel('Time [seconds]');
  ylabel('Position [radians]');
  text(7, 2, 'Unstable');
  SetLabels(12, 12,1)
  print('-deps2c', 'PIcontrol_PosRamp')
```

Figure 12: Position response of velocity with PI control, system is type I, unstable, SSE: undefined.

4 Frequency response and tracking accuracy

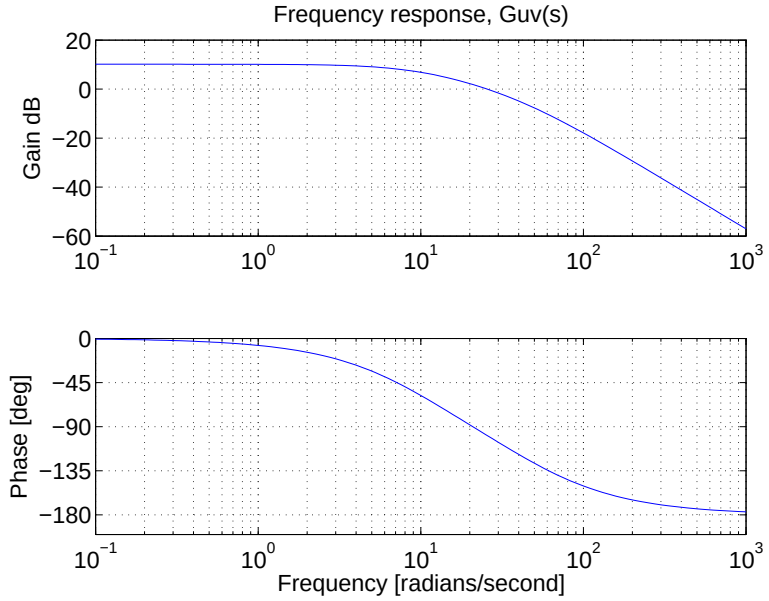
4.1 The frequency-response plot shows the value of a transfer function as a function of frequency

- X-Axis: frequency on a logrithmic axis, in [radians/second] or [cycles/second].
- First Y-Axis: magnitude in dB.

$$dB = 20 \log_{10} (|G(s = j\omega)|) \tag{18}$$

Gain	dB	Gain	dB	Gain	dB	Gain	dB
0	$-\infty$	0.5	-6 dB	1.0	0 dB	10	20
.1	-20 dB	0.707	-3 dB	2	6 dB	100	40

- Second Y-Axis: phase angle in degrees

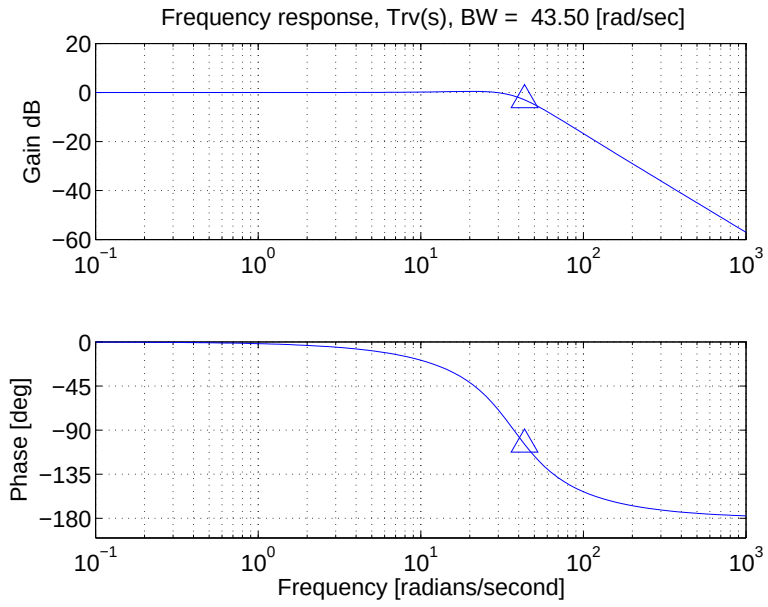



```
figure(1), clf
 bode(Guv)
 title('Frequency response, Guv(
 SetLabels(12); grid on
 print('-deps2c', 'OL_VBode')
```

Figure 13: Frequency response plot of $G_{uv}(s)$. DC gain = 1402/436.2 = 10.1 dB.

4.1.1 Bandwidth and example frequency-response plots

- Bandwidth: frequency where the magnitude curve falls off by 3 dB from its DC value.



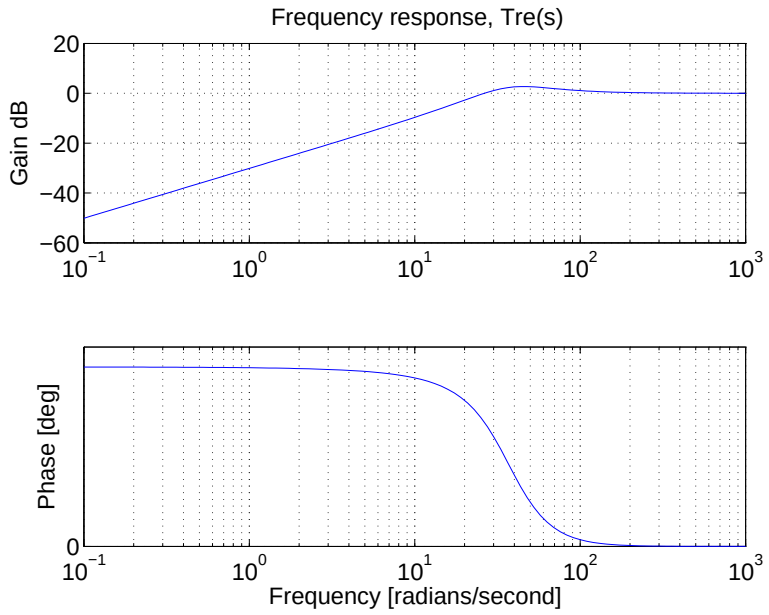

```
wBW = bandwidth(Trv)
[MagBW, PhaseBW] = bode(Trv, wBW
MagBWdB = 20*log10(MagBW)
figure(2), clf
 bode(Trv, {0.1, 1000})
 hs = findobj('type', 'axes')
 axes(hs(end)),
 hold on;  plot(wBW, MagBWdB, '^
 axes(hs(end-1)), hold on;
  plot(wBW, PhaseBW, '^');
 hold off
 title(sprintf('Frequency respon
 SetLabels(12); grid on
 print('-deps2c', 'Trv_FreqResp'
```

Δ: Point on the frequency response curve giving the closed-loop bandwidth.

Figure 14: Frequency response plot of $T_{rv}(s)$ with velocity PI control. $\omega_{BW} = 43$ [radians/second].

4.1.2 Tracking accuracy as a function of frequency

- The steady-state error specifies the error in steady-state, but some control systems never reach steady state.
  - Example: robot trajectory tracking for spray painting.
  - Tracking system in Lab part 7.
- For these applications, we need to look at the tracking error transfer function across a range of frequencies.
  - $T_{re}(s)$ evaluated at $s = j\omega$ gives the gain and phase of the error response.




```
Tre = feedback(1, Gc*Guv)
figure(3), clf
 bode(Tre, {0.1, 1000})
 title('Frequency response, Tre(
 SetLabels(12); grid on
 print('-deps2c', 'Tre_FreqResp'
```

Figure 15: Frequency response plot of tracking error $T_{re}(s)$ with velocity PI control. System shows tracking errors less than -20 dB up to 5 [radians/second].

5 Sensitivity

- The sensitivity of a system is the percentage change in the system response, per percent change in an element

$$\mathcal{S}\left(s=j\omega\right)=\frac{\%\,\text{Change in system transfer function}}{\%\,\text{Change in system component transfer function}} \tag{19}$$

  - In general, the sensitivity is a function of frequency
  - For an open-loop system, the sensitivity is 100 %
    (10% change in $G_p\left(s\right)$ gives 10% change in $G_{ry}\left(s\right)$)

- Closed-loop control can dramatically reduce the sensitivity. Franklin *et al.* in section 4.1.4 (7th ed.) work out the sensitivity in $T_{ry}\left(s\right)$ for the control loop of figure 16. They show:

$$\mathcal{S}_{G_p}^{T_{ry}}\left(s\right)=\frac{1}{1+G_c\left(s\right)G_p\left(s\right)H_y\left(s\right)} \tag{20}$$

  (Compare Eqn (20) with Franklin *et al.* Eqn (4.22) (7th ed.))

- It turns out that the sensitivity function is equal to the error response transfer function

$$\mathcal{S}_{G_p}^{T_{ry}}\left(s\right)=T_{re}\left(s\right) \tag{21}$$

- Looking back to figure 15, the PI control considered reduces the sensitivity of the closed-loop response to -34 dB below ω =1.0 [rad/sec]. -34 dB is a 50x reduction.

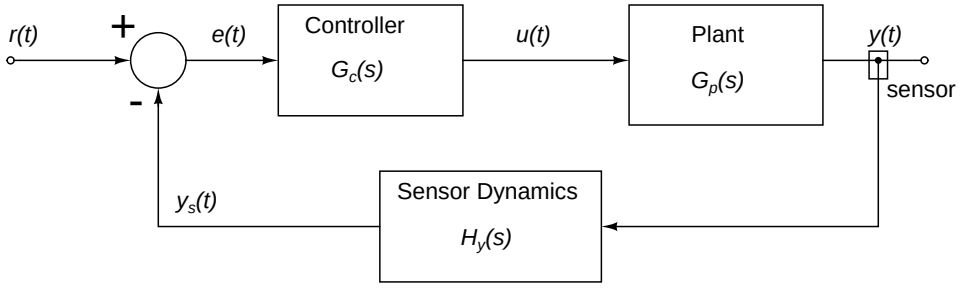


Figure 16: Standard block diagram.

6 Conclusions

- We have seen additional performance measures for feedback control systems:
  - System type and steady-state error
  - Tracking accuracy
  - Closed-loop bandwidth
  - Sensitivity
- Table 6 lists performance measures calculated for the example system.

Measure	Open-loop	P-type control $k_p = 1.0$	PI-type control $k_p = 1.0\,, k_i = 10.0$	Units
System type	0	0	I	
Steady-state error (step)	-	24%	0%	Percent
Steady-state error (ramp)	∞	∞	3.1%	Percent
ω_{BW} , bandwidth	9.4	47.4	43.5	[rad/sec]
Sensitivity, $\omega = 1.0$ [rad/sec]	100%	24%	3%	Percent

Table 6: Tabulation of performance of open-loop, p-type and PI-type velocity control of a motor drive.

- We have also seen that the PI control applied incautiously to the position servo loop leads to instability, showing the necessity of a thorough feedback-control design method.